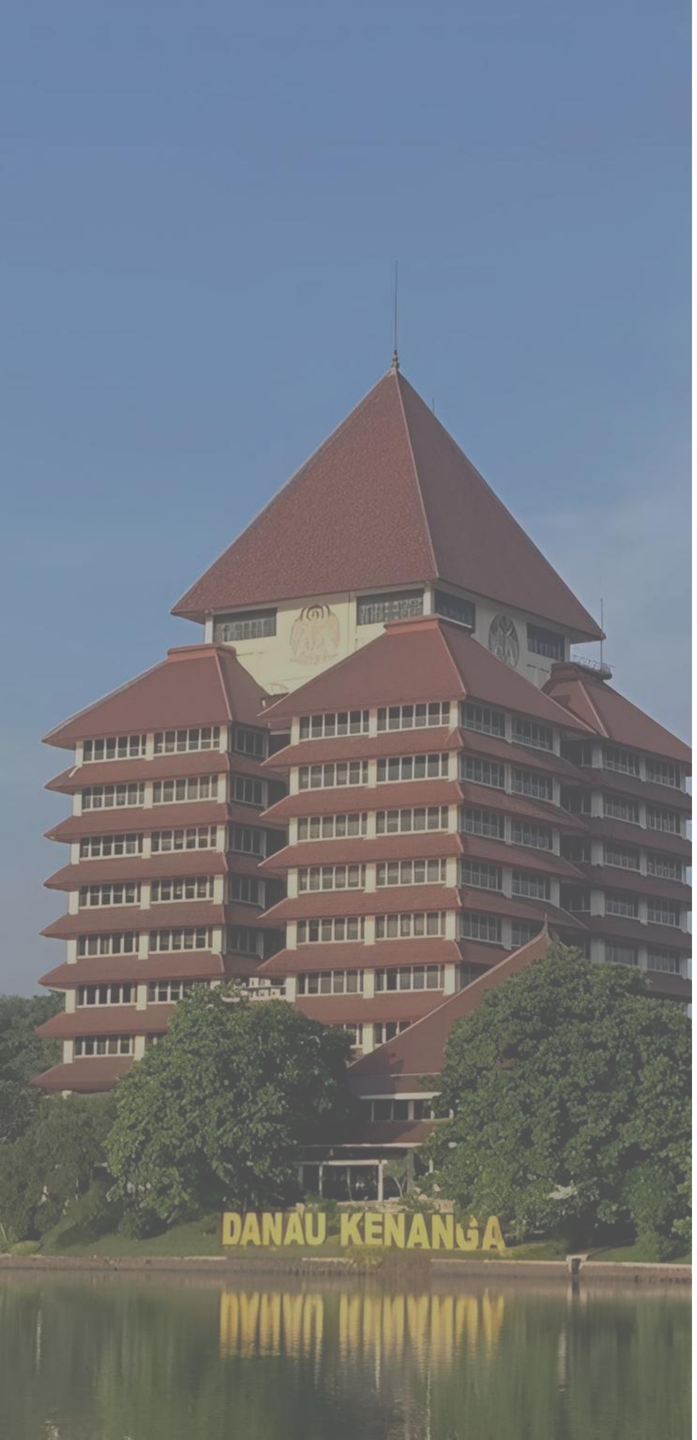

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

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Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Pendahuluan

Latar Belakang Penelitian (1/2)

Otak merupakan sistem saraf pusat manusia. **Kanker otak** merupakan masalah kesehatan serius dengan **tingkat kematian yang tinggi.**

Pengobatan utama kanker otak adalah **radioterapi**. Teknik yang saat ini sangat berkembang adalah **Volumetric Modulated Arc Therapy (VMAT)**.

Data GLOBOCAN (2022)

321.731
Kasus baru global

248.500
Angka kematian

Proyeksi masa depan (2045)

↑47%
Peningkatan mencapai **474.000** kasus baru

(Filho et al., 2025)

Pendahuluan

Latar Belakang Penelitian (2/2)

Keunggulan VMAT

- Menghasilkan distribusi dosis yang presisi pada target.
- Durasi penyinaran lebih singkat.

VMAT membutuhkan TPS

PROBLEM

Treatment Planning System (TPS)

- Kualitas perencanaan radioterapi bergantung keterampilan dan pengalaman perencana.
- Kompleks dan memerlukan waktu cukup panjang.

SOLUTION

Machine Learning

- Mengurangi waktu perencanaan radioterapi.
- Meminimalisir human error.

Pendekatan machine learning diharapkan dapat **meningkatkan efisiensi** dan **kualitas** perencanaan radioterapi **tanpa mengorbankan akurasi** dan **keamanan** pasien.

Pendahuluan

Rumusan Masalah dan Tujuan Penelitian

Rumusan Masalah

- Bagaimana performa model XGBoost dalam memprediksi distribusi dosis mencakup D2%, D50%, D98% pada PTV, serta Dmax dan Dmean pada OAR untuk perencanaan VMAT kanker otak?
- Bagaimana perbandingan kualitas perencanaan radioterapi berdasarkan Homogeneity Index (HI) dan Conformity Index (CI) antara hasil prediksi model XGBoost dengan perencanaan radioterapi klinis?

Tujuan Penelitian

- Mengevaluasi performa model XGBoost dalam memprediksi distribusi dosis perencanaan radioterapi mencakup D2%, D50%, D98% pada PTV, serta Dmax dan Dmean pada OAR untuk perencanaan VMAT kanker otak.
- Menganalisis dan membandingkan kualitas perencanaan radioterapi berdasarkan Homogeneity Index (HI) dan Conformity Index (CI) antara hasil prediksi model XGBoost dan perencanaan radioterapi klinis.

Pendahuluan

Batasan Masalah dan Manfaat Penelitian

Batasan Masalah

- Data yang digunakan adalah data pasien kanker otak dengan teknik radioterapi VMAT dari MRCCC Siloam Hospitals Semanggi.
- Model machine learning yang digunakan adalah XGBoost.
- OAR yang ditinjau terbatas pada organ kritis utama pada kanker otak, yaitu batang otak (brainstem), saraf optik kanan dan kiri, mata kanan dan kiri, serta lensa kanan dan kiri.
- Parameter dosis yang dievaluasi adalah D2%, D50%, D98% pada PTV, serta dosis maksimum (Dmax) dan dosis rata-rata (Dmean) pada OAR.

Manfaat Penelitian

Melalui penelitian ini, diharapkan dapat meningkatkan wawasan terkait prediksi distribusi dosis menggunakan algoritma XGBoost, serta berpotensi diterapkan secara klinis untuk mendukung pengembangan perencanaan radioterapi otomatis yang lebih efisien dan akurat.

Tinjauan Pustaka

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Tinjauan Pustaka

Radioterapi: Volumetric Modulated Arc Therapy (VMAT)

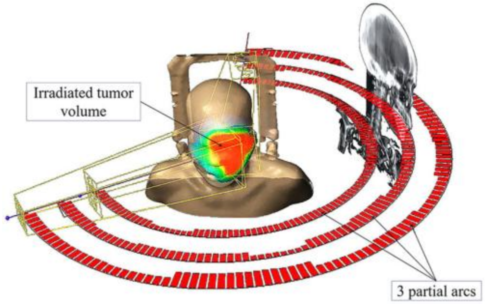
Kanker otak merupakan kondisi ketika terjadi pertumbuhan sel secara abnormal dalam jaringan otak.

Tujuan radioterapi menghancurkan sel kanker dan meminimalkan kerusakan organ sehat.

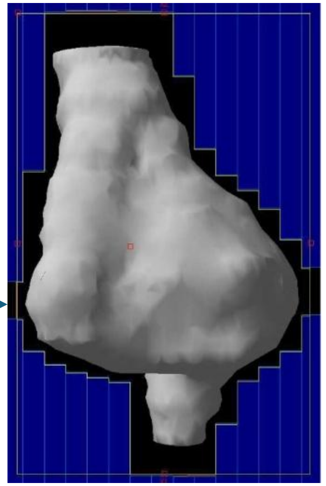
Linear Accelerator (LINAC)

LINAC memancarkan radiasi energi tinggi dari luar tubuh menuju kanker dengan presisi dan akurasi tinggi (radioterapi eksternal).

Volumetric Modulated Arc Therapy (VMAT)



Kecepatan rotasi gantry, laju dosis, dan bentuk multileaf collimator (MLC) bervariasi dengan penyinaran berkas radiasi tetap menyala.

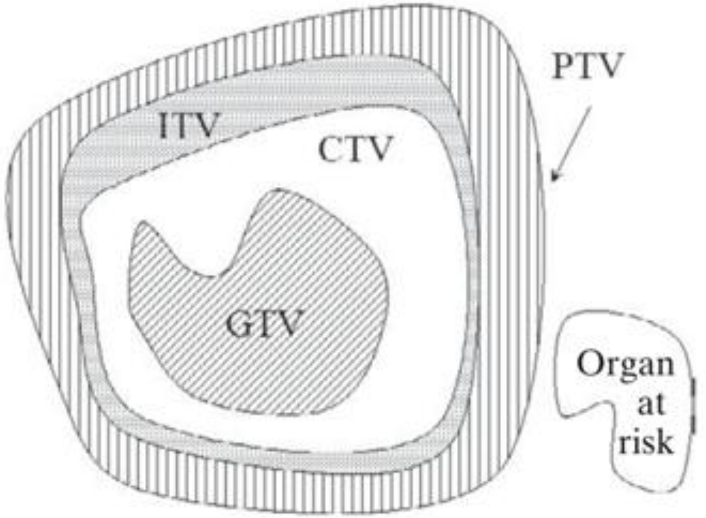


Multileaf collimator (MLC)
(Taylor & Powel, 2004)

Tinjauan Pustaka

Perencanaan Radioterapi

Simulasi/ Pencitraan → Konturing (Delineasi Volume Target) → Optimasi Dosis di TPS → Quality Assurance (QA)

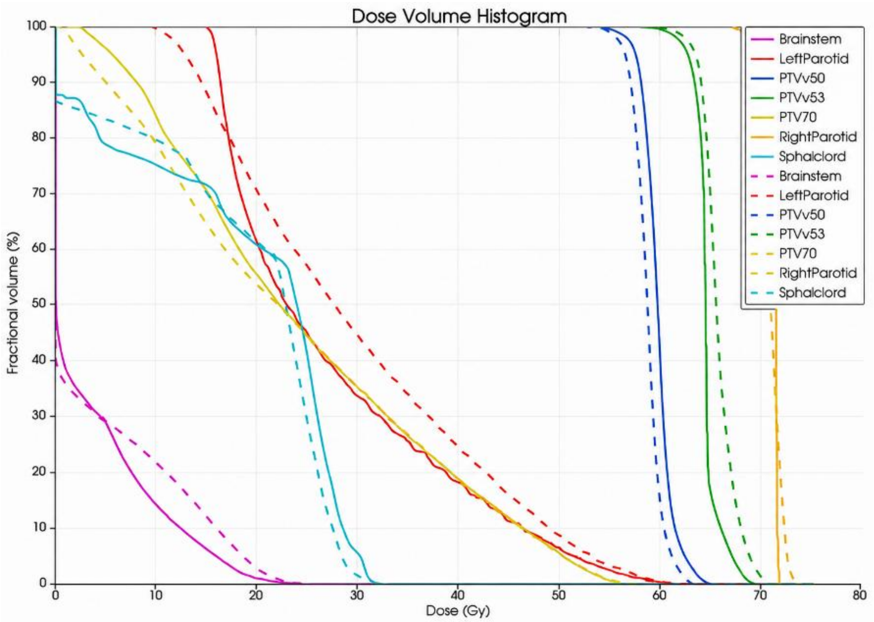


GTV: Gross Target Volume
CTV: Clinical Target Volume
ITV: Internal Target Volume
PTV: Planning Target Volume
OAR: Organ at Risk

Representasi grafik volume target (Podgorsak, 2005)

Tinjauan Pustaka

Evaluasi Perencanaan Radioterapi



Kurva DVH dengan data aktual (garis solid) dan nilai prediksi (garis putus putus) (Chandran et al., 2023)

Dose Volume Histogram (DVH) merupakan keluaran TPS yang digunakan untuk **mengevaluasi kesesuaian distribusi dosis radiasi** pada target dan organ sehat.

Homogeneity Index (HI)

- Mengukur tingkat keseragaman distribusi dosis pada PTV.
- HI mendekati 0 → distribusi dosis semakin homogen.

Conformity Index (CI)

- Mengevaluasi kesesuaian distribusi dosis yang diberikan pada PTV.
- CI mendekati 1 → konformitas dosis semakin optimal.

Tinjauan Pustaka

Fitur Radiomik dan Dosiomik

Aspek	Radiomik	Dosiomik
Sumber data	ROI (region of interest) yang digambar dokter onkolog pada citra medis	Kurva DVH
Informasi yang diekstraksi	• Bentuk & struktur anatomi • Kondisi tekstur patologis	• Keseragaman dosis • Variasi distribusi dosis
Contoh	Elongation, flatness, least axis length, mesh volume	Dmean, Dmax, Dmin, D2%, D50%, D98%

Tinjauan Pustaka

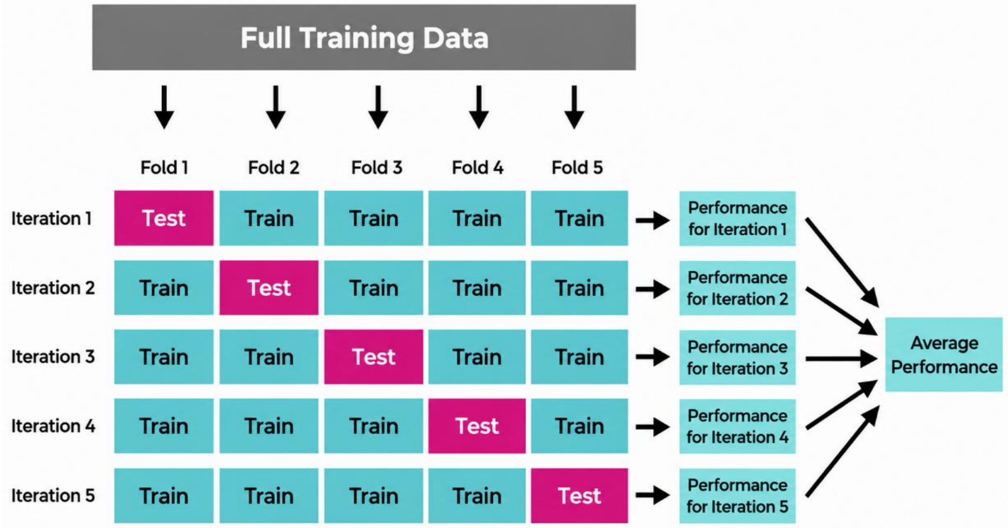
XGBoost dan K-Fold Cross Validation

Prinsip Kerja XGBoost

Ensemble learning menggabungkan beberapa **decision tree** secara bertahap untuk memperbaiki kesalahan prediksi sebelumnya.

Keunggulan XGBoost

- Cocok untuk data tabular.
- Tahan overfitting dengan regularisasi.
- Fleksibel dalam penyesuaian parameter model.
- Efisien secara komputasi



Arsitektur K-fold cross validation (Lumumba et al., 2024)

K-Fold CV membagi dataset menjadi beberapa bagian secara bergantian sebagai data latih dan data uji.

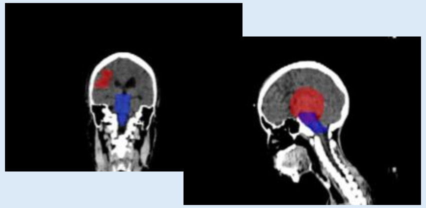
Metode Penelitian

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Metode Penelitian

Preprocessing Data

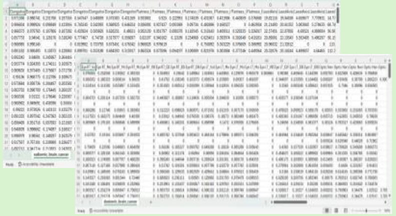
Pengumpulan Data → **Ekstraksi Data** → **Normalisasi Data** → **Seleksi Target**



Data DICOM 178 pasien kanker otak

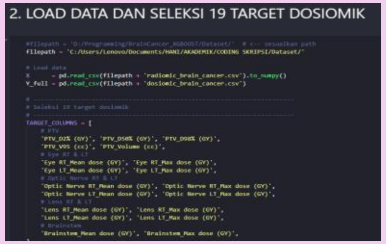
Hasil ekstraksi:

- 14 fitur radiomik
- 64 fitur dosiomik



Fitur dosiomik dibagi dengan dosis preskripsi pasien agar berskala seragam (0–1)

$$D_{norm} = \frac{D_x}{D_{preskripsi}}$$



Seleksi 19 fitur dosiomik sebagai target:

- PTV: D2%, D50%, D98%, V95%, VPTV
- OAR: Dmax & Dmean

Metode Penelitian

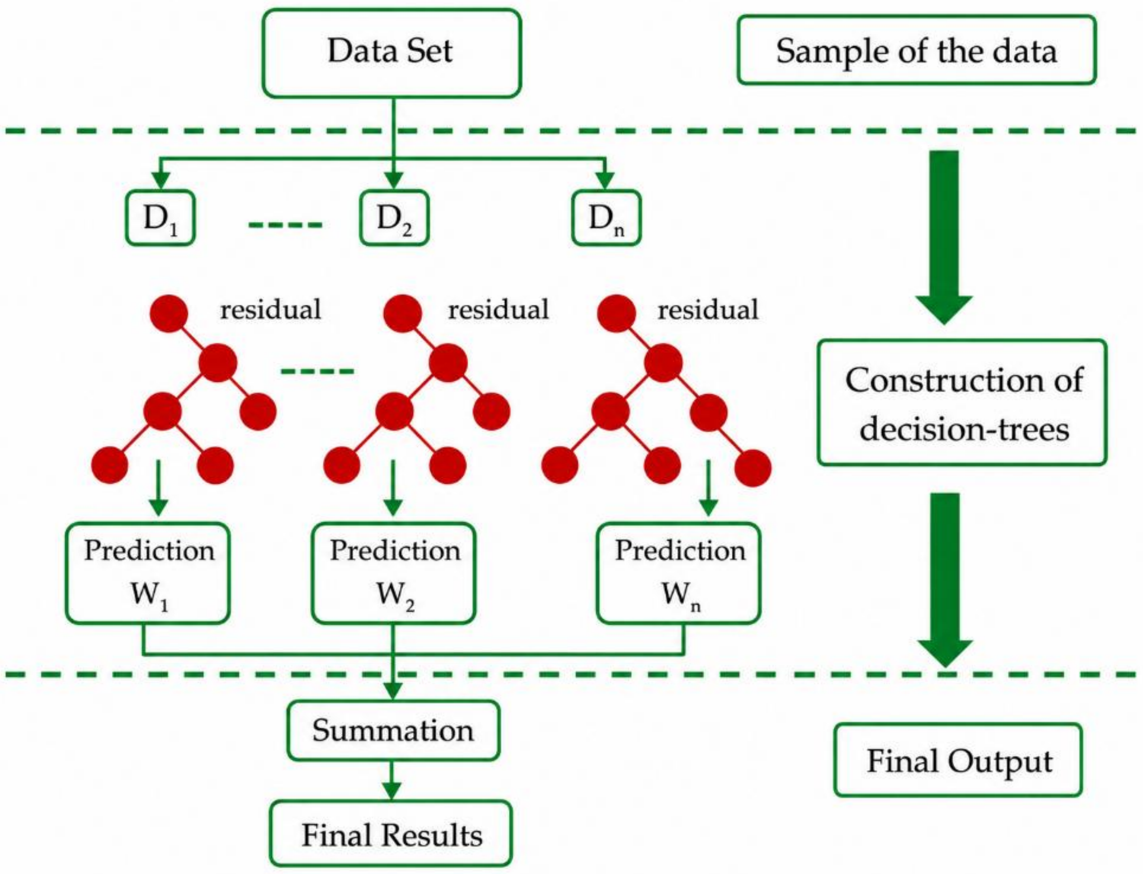
Pembangunan Model XGBoost

Pembagian Dataset

- Training: 70% (124 data)
- Testing: 30% (54 data)

Hyperparameter Tuning

GridSearchCV diintegrasikan dengan K-Fold CV (k=5)



Arsitektur XGBoost (Khan et al., 2022)

Metode Penelitian

Metrik Evaluasi

Homogeneity Index (HI)

$$HI = \frac{(D_{2\%} - D_{98\%})}{D_{50\%}}$$

D2% = dosis yang diterima oleh 2% volume kanker
D50% = dosis yang diterima oleh 50% volume kanker
D98% = dosis yang diterima oleh 98% volume kanker

Rumus HI berdasarkan ICRU 83 (Su et al., 2025)

Conformity Index (CI)

$$CI = \frac{V_{RI}}{TV}$$

VRI = volume target yang menerima 95% dosis yang diresepkan
TV = volume total dari target tumor

Rumus CI berdasarkan RTOG (Adeneye et al., 2021)

Mean Square Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

y_i = nilai sebenarnya
$\hat{y}_i$ = nilai yang diprediksi
n = jumlah data

(Airlangga, 2024)

Uji Wilcoxon

$$W = \min(W^+, W^-)$$

W^+ = Jumlah ranking positif
W^- = Jumlah ranking negatif
p>0,05 $\rightarrow$ tidak terdapat perbedaan signifikan terhadap data klinis dan prediksi

(Ramlah et al., 2023)

Hasil dan Pembahasan

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Hasil

Parameter Optimal dan Perbandingan Performa Model

Parameter optimal model XGBoost

Parameter	Hasil
colsample_bytree	0,3
max_depth	2
n_estimators	50

Tabel perbandingan performa model XGBoost dan Random Forest

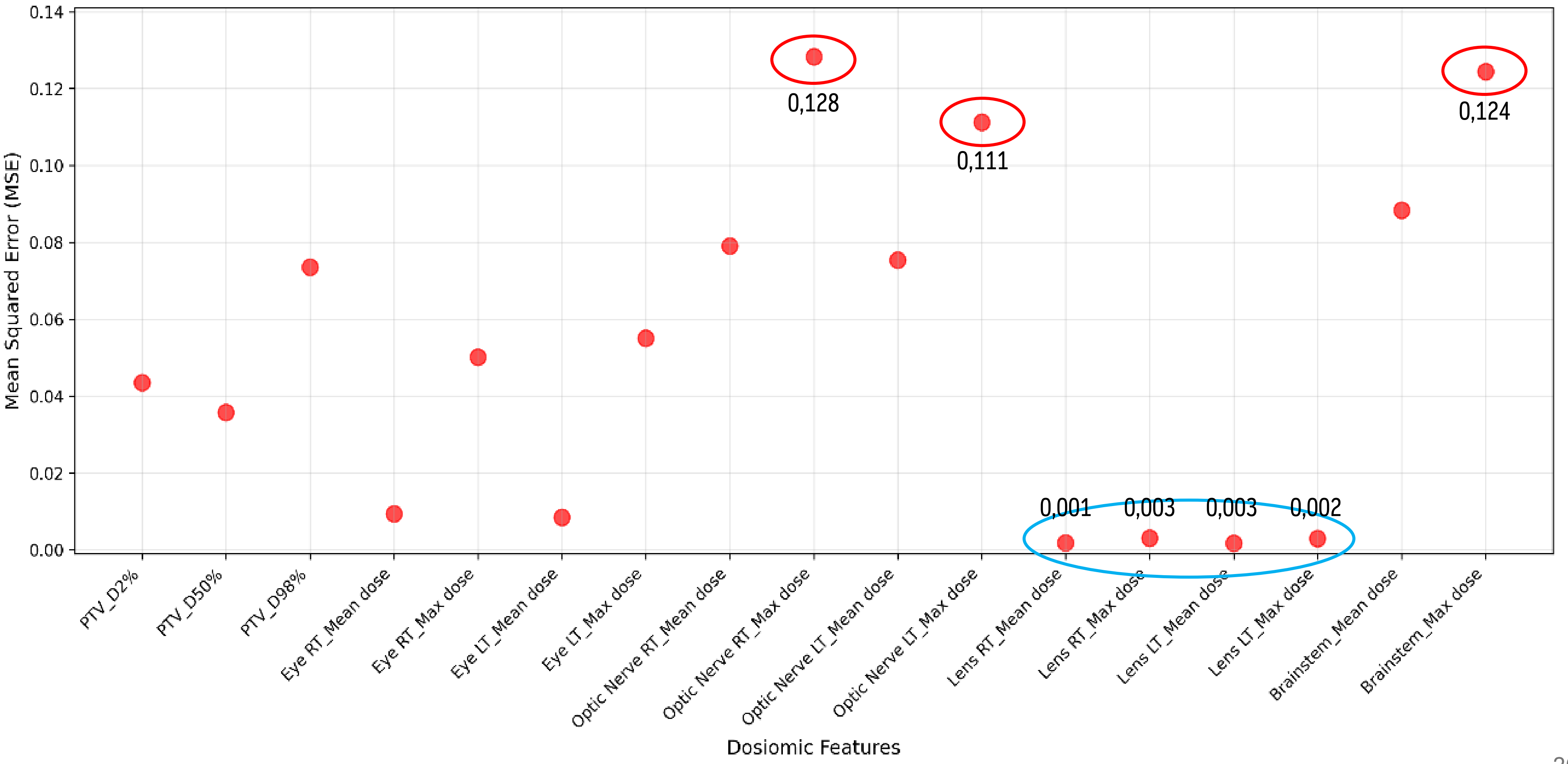
Model	Seleksi Target	Jumlah Target	MSE	Waktu Pelatihan	Uji Wilcoxon Per Fitur (p>0,05)
XGBoost	**Ya**	**19**	**0,047**	**53 detik**	**19/19**
	Tidak	19	0,047	4 menit 30 detik	16/19
		64	0,040		
Random Forest (Azwat, 2024)	Ya	19	0,049	13 menit 55 detik	15/19
	Tidak	19	0,049	16 menit 22 detik	15/19
		64	0,039		

XGBoost dengan seleksi fitur unggul pada seluruh metrik evaluasi karena tidak terganggu oleh fitur yang tidak relevan dan mekanisme gradient boosting mampu memperbaiki kesalahan prediksi secara bertahap.

Hasil

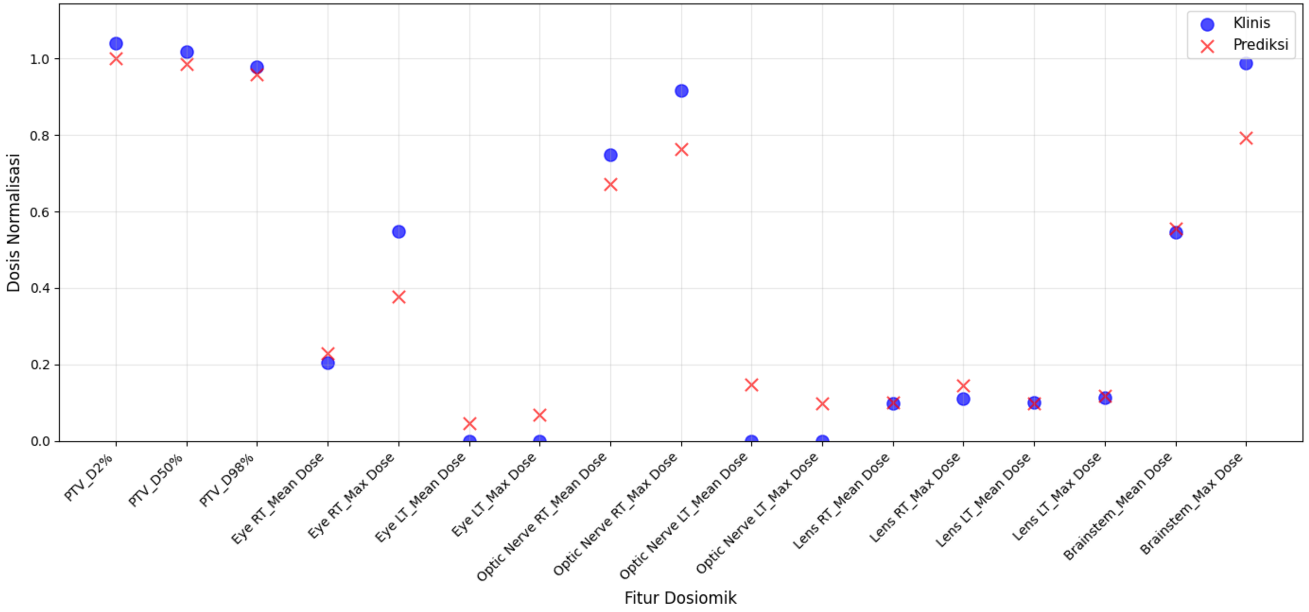
Performa Model

Hasil MSE pada setiap fitur yang dievaluasi

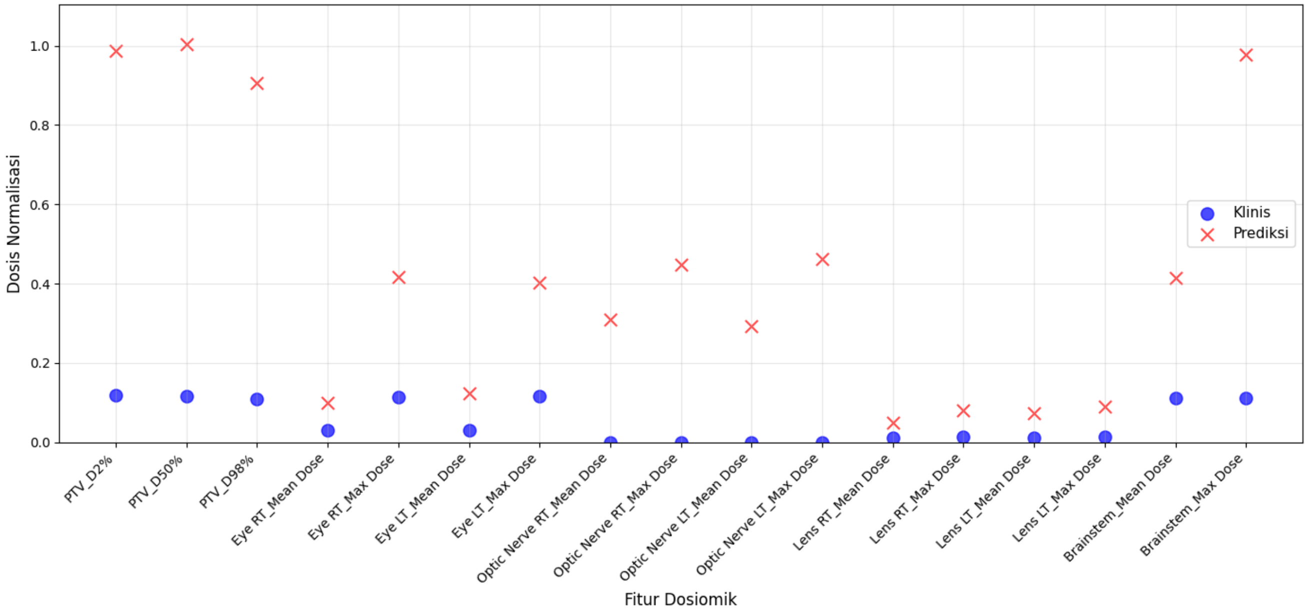


Hasil

Performa Model



Hasil prediksi dosis pada pasien 5 dengan error terkecil

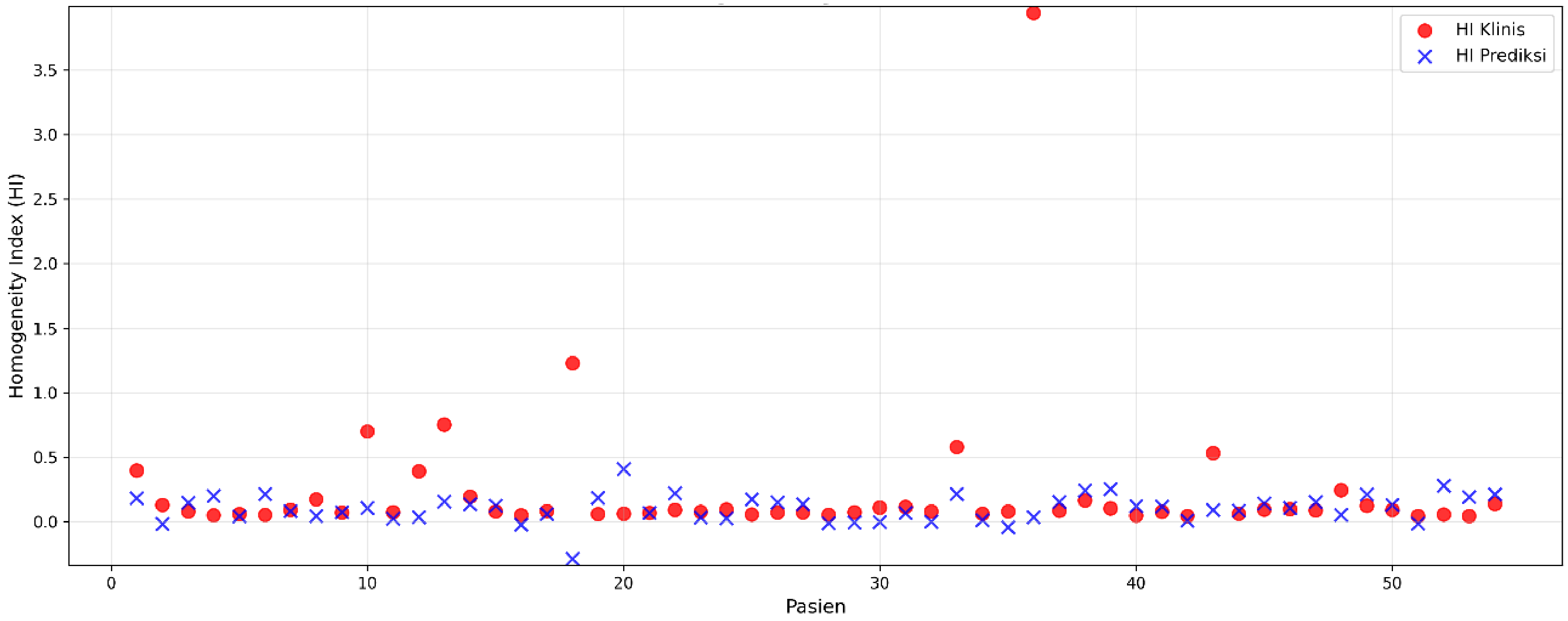


Hasil prediksi dosis pada pasien 7 dengan error terbesar

Hasil

Evaluasi Homogenity Index (HI)

Perbandingan nilai HI klinis dan prediksi

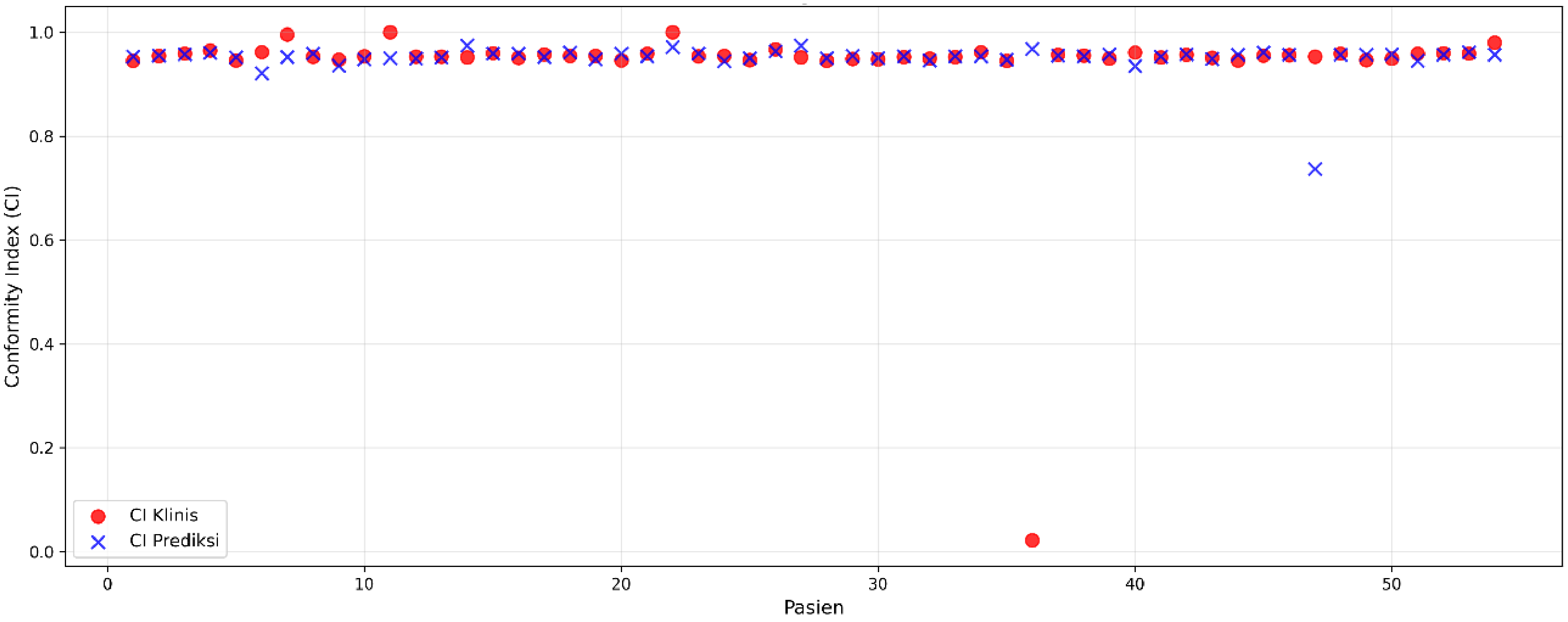


- HI klinis = 0,232 ± 0,553
- HI prediksi = 0,102 ± 0,106
- *p-value* = 0,425
- MSE = 0,357

Hasil

Evaluasi Conformity Index (CI)

Perbandingan nilai CI klinis dan prediksi



- CI klinis = 0,939 ± 0,126
- CI prediksi = 0,950 ± 0,030
- *p-value* = 0,766
- MSE = 0,017

Hasil

Uji Statistik

Tabel hasil uji statistik pada PTV dan OAR

Feature	Clinical (mean ± SD)	Predicted (mean ± SD)	P-value
PTV_D2%	0,976 ± 0,213	0,960 ± 0,132	0,061
PTV_D50%	0,927 ± 0,224	0,925 ± 0,124	0,087
PTV_D98%	0,817 ± 0,279	0,868 ± 0,141	0,853
Eye RT_Mean dose	0,139 ± 0,091	0,148 ± 0,060	0,510
Eye RT_Max dose	0,342 ± 0,227	0,344 ± 0,129	0,721
Eye LT_Mean dose	0,138 ± 0,096	0,153 ± 0,066	0,186
Eye LT_Max dose	0,333 ± 0,232	0,352 ± 0,135	0,431
Optic Nerve RT_Mean dose	0,387 ± 0,278	0,363 ± 0,150	0,573
Optic Nerve RT_Max dose	0,541 ± 0,356	0,538 ± 0,195	0,955
Optic Nerve LT_Mean dose	0,342 ± 0,267	0,375 ± 0,174	0,295
Optic Nerve LT_Max dose	0,495 ± 0,348	0,482 ± 0,214	0,840
Lens RT_Mean dose	0,079 ± 0,044	0,081 ± 0,033	0,727
Lens RT_Max dose	0,096 ± 0,056	0,102 ± 0,037	0,392
Lens LT_Mean dose	0,083 ± 0,045	0,079 ± 0,030	0,596
Lens LT_Max dose	0,102 ± 0,059	0,102 ± 0,037	0,734
Brainstem_Mean dose	0,513 ± 0,317	0,520 ± 0,211	0,990

Kesimpulan

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Kesimpulan

1. Model **XGBoost dengan seleksi fitur berhasil memprediksi distribusi dosis** (D2%, D50%, D98% pada PTV; Dmax & Dmean pada OAR) dengan rata-rata **MSE = 0,047** dalam waktu pelatihan **53 detik.**
   - Akurasi tertinggi pada organ Lens RT_Mean Dose (MSE = 0,001) dan error tertinggi pada Optic Nerve RT_Max dose (MSE = 0,128).
   - Uji Wilcoxon: seluruh 19 fitur memperoleh $p > 0{,}05 \rightarrow$ tidak ada perbedaan signifikan antara nilai klinis dan prediksi.

2. Kualitas perencanaan radioterapi memiliki nilai prediksi yang sebanding dengan nilai klinis.
   - **HI prediksi: 0,102 ± 0,106** vs. HI klinis: 0,232 ± 0,553; p-value HI = 0,425
   - **CI prediksi: 0,950 ± 0,030** vs. CI klinis: 0,939 ± 0,126; p-value CI = 0,766

Saran

Penelitian selanjutnya disarankan untuk menggunakan data yang lebih besar dan beragam untuk meningkatkan kemampuan generalisasi model. Preprocessing kontur OAR yang tumpang tindih dengan PTV perlu dilakukan lebih teliti sebelum pelatihan model, serta eksplorasi algoritma atau model machine learning lainnya dapat dilakukan untuk membandingkan performa dengan model XGBoost.

Referensi

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

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Terima Kasih

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Slide Tambahan

Prediksi Distribusi Dosis pada Radioterapi Teknik Volumetric Modulated Arc Therapy (VMAT) untuk Kanker Otak Menggunakan Algoritma Extreme Gradient Boost (XGBoost)

Metode Penelitian

Fitur Dosiomik

No.	Fitur Dosiomik (PTV)
1	**PTV Volume**
2	**PTV V95**
3	**PTV D2%**
4	**PTV D50%**
5	**PTV D98%**
6	PTV Mean Dose
7	PTV Min Dose
8	PTV Max Dose

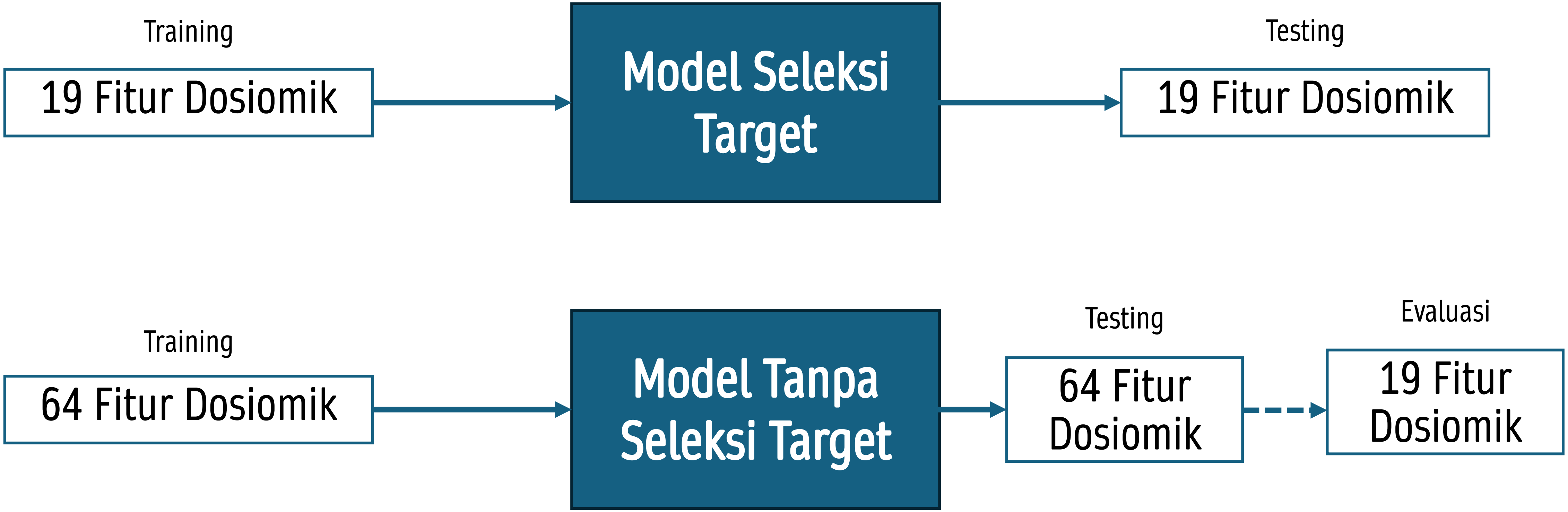
No.	Fitur Dosiomik (OAR)
1	**Eye RT Mean Dose**
2	**Eye RT Max Dose**
3	Eye RT Min Dose
4	Eye RT Volume
5	Eye RT V95
6	Eye RT D2%
7	Eye RT D50%
8	Eye RT D98%

- **7 OAR** x 8 fitur dosiomik = 56 fitur dosiomik OAR
- 56 fitur dosiomik OAR + 8 fitur dosiomik PTV = 64 fitur dosiomik

Total fitur dosiomik = 64 fitur dosiomik

Metode Penelitian

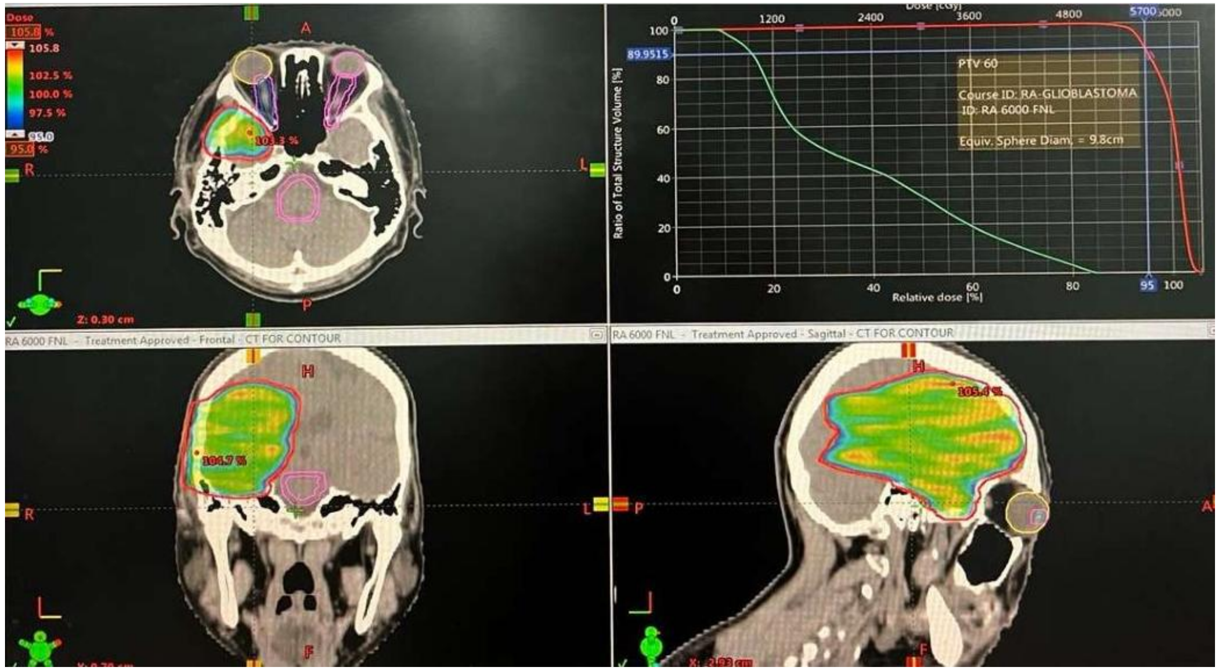
Model Fitur Dosiomik

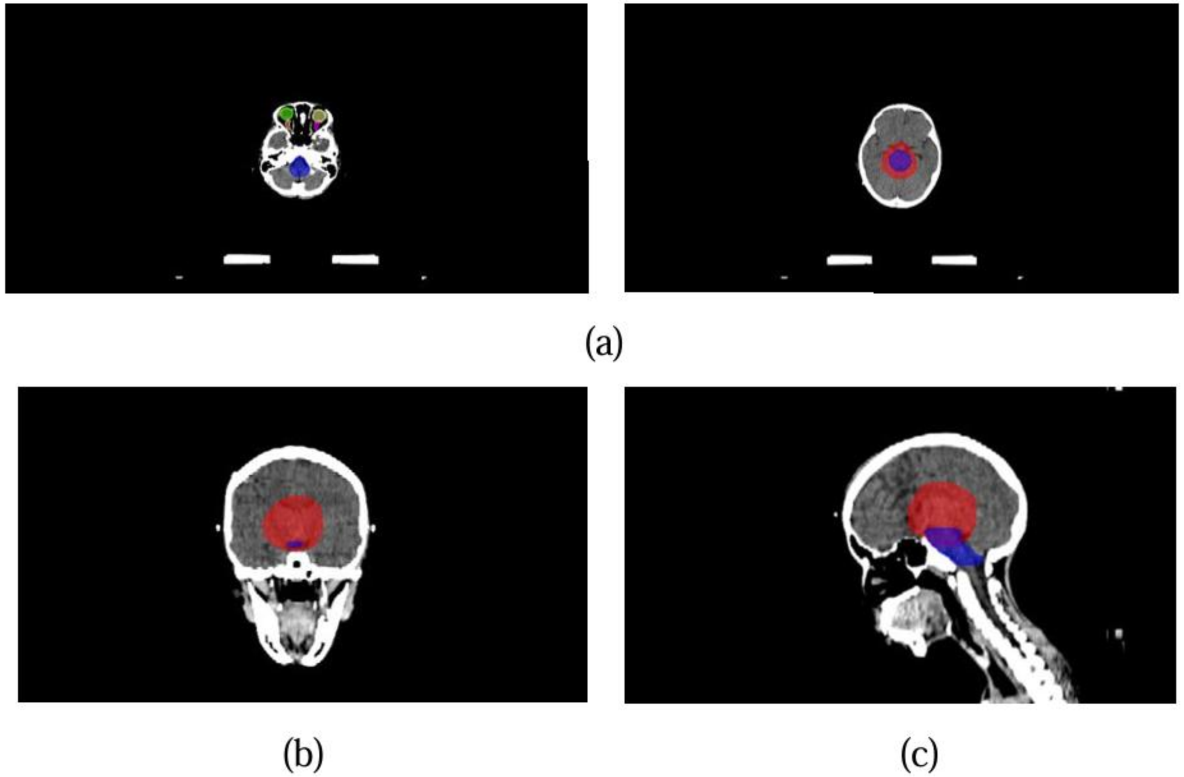


Hasil

Data DICOM

Kontur otak. (a) Axial (b) Coronal (c) Sagittal





Hasil

CI Training

Hasil

HI Training

Plot HI data training